

# PHILIPPE FANARO

## SOFTWARE ENGINEER

NEXT.JS & REACT.JS SPECIALIST | TYPESCRIPT & AI

### 1 Summary

I build fast. With an AI-powered workflow, I deliver beautiful frontend experiences alongside a performant backend.

Software Engineer with **3+ years** of experience building full-stack, cloud-native, and distributed software systems across the complete software development lifecycle. I design and deliver scalable SaaS platforms, APIs, and high-performance web applications using **Next.js**, **TypeScript**, **React.js** and Cloud Services, combining frontend architecture expertise with backend engineering, cloud infrastructure, data modeling, and system design to build reliable, maintainable, and production-grade products.

Besides Next.js and React.js, I have experience working with **Flutter** and **Dart**, having built a popular open source data structures package (200+ Stars on Github). I have also built frontend products in the insurance sector, consisting of Web Components which integrated seamlessly with third-party platforms. I am a programmer-writer as well, having written two books using **L<sup>A</sup>T<sub>E</sub>X**, while at the same time using that same programming language to create scalable vector diagrams.

Check out my portfolio and blog: **neofanaroio.vercel.app**.

## 2 Work Experience

### 1. **Programmer and Writer** (*Sep 2024 – Dec 2024*)

- Designed and developed an open-source  $\text{\LaTeX}$  package for generating vector Go (Baduk/Weiqi) board diagrams, and used it to write and typeset a complete book, “*Técnicas de Go.*”
- Built a reusable  $\text{\LaTeX}$ /TeX package that renders scalable vector graphics (SVG/PDF-quality) Go diagrams from code, replacing manual image creation.
- Authored and typeset a 400-page technical book in Portuguese using  $\text{\LaTeX}$ , covering tesujis, life-and-death, and other techniques.
- Designed a clean, documented API so other authors can produce publication-quality diagrams with minimal setup.
- Managed the full project lifecycle independently: requirements, software design, implementation, documentation, version control (Git/GitHub), and release.
- Published under an open-source license on GitHub (20+ Stars): [github.com/psygo/tecnicas\\_de\\_go](https://github.com/psygo/tecnicas_de_go)

### 2. **Full Stack Developer** @ Vertical Insure, Inc. (*Feb 2023 – Jun 2023*) – *United States (Remote)*

- Full Stack Developer working across the entire application stack, from frontend development to backend services, cloud infrastructure, and observability.
- Built and maintained web applications using React, Lit Framework, JavaScript, and TypeScript, with a focus on reusable components and responsive user interfaces.
- Developed backend services and APIs using Java, Micronaut, Node.js, and Express, integrating application logic with external APIs from the insurance sector.

- Worked across the full development lifecycle, from implementing new features and debugging existing systems to deployment and production monitoring.
- Managed cloud and deployment infrastructure using AWS and Render, with Datadog for monitoring and observability.
- Used tools such as Retool to rapidly develop internal interfaces and operational tools.
- Collaborated across the frontend, backend, and infrastructure layers, gaining experience in designing and connecting the different components of a production software system.
- Tech Stack:
  - Frontend: React, Lit Framework, JS | TS
  - Backend: Java, Micronaut, NodeJS, Express
  - DevOps: AWS, Render, DataDog, Retool, etc.

### 3. **Open Source Software Developer – Dart / Flutter** @ Freelance (*Sep 2020 – Jan 2021*) – *Online*

- Co-developed an open source data structure package for Dart/Flutter (Fast Immutable Collections, or FIC):  
[github.com/marcglassberg/fast\\_immutable\\_collections](https://github.com/marcglassberg/fast_immutable_collections)
- Designed and implemented immutable lists, sets, maps, and multimaps, as well as extension methods, collection utilities, and custom comparators for native Dart collections.
- Worked on data structures, algorithms, performance optimization, API design, package development, testing, documentation, and Git-based collaborative software development.
- Contributed to the design and implementation of collection types optimized to provide performance comparable to native mutable Dart collections while preserving immutability.

- Collaborated with other developers through GitHub, Git, code review, issue tracking, and Kanban-based development workflows.

#### 4. **Data Scientist @ Zanthus** (*Feb 2019 – Oct 2019*) – *Greater São Paulo Area*

- Full-stack data scientist at Zanthus, a retail automation software company, owning the complete data pipeline from extraction to production API.
- Built a Customer Lifetime Value (CLV) model to predict customer value for retail clients, supporting marketing, retention and segmentation decisions.
- Extracted and queried large retail transaction datasets using SQL from PostgreSQL and Oracle databases.
- Performed data cleaning, preprocessing, and feature engineering in Python using pandas and NumPy.
- Developed and evaluated machine learning and statistical models with scikit-learn and lifetimes.
- Designed and deployed a REST API with Django to serve model predictions and data to internal systems.
- Delivered end-to-end data science projects independently: data extraction, ETL, modeling, deployment, and stakeholder communication.
- Tech Stack: Python, Pandas, NumPy, Scikit-learn, Django, and SQL.

#### 5. **Intern @ Vital Care App** (*Sep 2016 – Dec 2016*) – *Greater São Paulo Area*

- Improved the customer service, digital marketing, and SEO strategies.

- Proposed and implemented user interface (UI) and user experience (UX) enhancements to the mobile app.
- Investigated regulatory compliance issues with partner pharmacies, analyzing health and sanitary infractions to help protect marketplace quality and customer trust.
- Worked across product, marketing, and operations in a fast-paced early-stage startup environment.

### 3 Education

1. **Telecommunications Engineer, Electrical Engineering (BSc) @ Universidade de São Paulo (USP) (*2011 – 2016*)**
  - The best university in South America.
  - Specializing in telecommunications.
  - (Engineering degrees take 5 years in Brazil.)
2. **International Engineering Exchange, Electrical Engineering @ Université Libre de Bruxelles (ULB) (*2015 – 2016*)**
  - One year abroad in Brussels, Belgium, studying engineering, through a government scholarship.
  - Perfected my English
  - Learned French.
3. **Masters (Latu Sensu), Psychology @ Myongji University (*2017*)**

### 4 Projects

- **Lang101** (*Jun 2026 – Today*)
  - A language learning app based on flashcards and Elo.

- Built with React Native and AI.
- **neofanaroio.vercel.app** (*Jun 2026 – Today*)
  - My personal portfolio and blog, built with Next.js and Tailwind-CSS.
  - The new version of my old, basic blog, fanaro.io.
  - Live: neofanaroio.vercel.app
- **Magi** (*Sep 2025 – Dec 2025*)
  - A 2D exploration of the internet, using SQL and Excalidraw to create a graph database.
  - [github.com/psygo/magi](https://github.com/psygo/magi)
  - Live: magi.vercel.app
- **Aquarifolio** (*Sep 2025 – Dec 2025*)
  - Interactive 3D portfolio built with React and Three.js showcasing software engineering projects.
  - [github.com/psygo/aquarifolio](https://github.com/psygo/aquarifolio)
  - Live: aquarifolio.vercel.app
- **Técnicas de Go (Go Board Game Techniques Book)** (*Sep 2024 – Nov 2024*)
  - A book on Go (board game), entirely made with L<sup>A</sup>T<sub>E</sub>X, and a custom package for drawing vector diagrams with TikZ/PGF.
  - [github.com/psygo/tecnicas\\_de\\_go](https://github.com/psygo/tecnicas_de_go)
- **YouTube KBD Nav** (*Sep 2021 – Oct 2021*)
  - A browser extension for navigating YouTube solely through the keyboard.

- [github.com/psygo/youtube\\_kbd\\_nav](https://github.com/psygo/youtube_kbd_nav)
- **Tradução de Livro de Go (Go Book Translation)** (*Aug 2021 – Sep 2021*)
  - Official translation and editing (L<sup>A</sup>T<sub>E</sub>X) of an introductory book on Go, from English to Portuguese.
  - [github.com/psygo/traducao\\_como\\_jogar\\_go](https://github.com/psygo/traducao_como_jogar_go)
- **fanaro.io** (*Aug 2021 – Sep 2021*)
  - My old blog, with no frameworks, HTML, CSS, and Web Components.
  - [github.com/psygo/fanaro.io](https://github.com/psygo/fanaro.io)
  - [fanaro.io](https://fanaro.io)

## 5 Certifications

- Deep Learning Specialization
- EF SET English Certificate 76/100 (C2 Proficient)
- Data Science Forum Participant

## 6 Publications

- **Técnicas de Go** – A 400-page technical book on Go (Baduk/Weiqi), written and typeset entirely in L<sup>A</sup>T<sub>E</sub>X.
- **Tradução de Livro de Go** – Official translation and editing of an introductory book on Go, from English to Portuguese.

## 7 Competencies

- Frontend Development
- Backend Development
- REST APIs
- Cloud Deployment
- Data Analysis
- Machine Learning
- UI Design and Development
- SQL Database Design
- Agile Development
- Test-Driven Development
- Code Reviewing

## 8 Technical Skills

- Top Skills (LinkedIn):
  - English
  - Writing
  - Artificial Intelligence (AI)
- Main Stack:
  - React
  - Next.js

- TypeScript
- SQL
- Tailwind CSS
- HTML
- CSS
- Other technologies I have worked with:
  - Flutter/Dart
  - Web Components
  - Lit (Web Components)
  - Three.js
  - React Native
  - Firebase
  - L<sup>A</sup>T<sub>E</sub>X
- Data Science/Analytics stack I have worked with:
  - Python
  - SQL
  - Pandas
  - NumPy
  - Scikit-learn
  - Django
- Other technologies I use or have used:
  - Java
  - Shell
  - Vim

- TikZ/PGF
- Git
- Github
- Docker
- REST APIs
- JSON
- Linux
- AWS

## 9 Languages

| Language   | Proficiency         |
|------------|---------------------|
| Portuguese | Native or Bilingual |
| English    | Native or Bilingual |
| French     | Advanced            |
| Spanish    | Intermediate        |
| Korean     | Basic               |
| Japanese   | Basic               |

## 10 Personal Interests

On my free time, I enjoy reading and watching videos about psychology, history, philosophy and economics.

I'm also a big fan of board games in general, not only of Go (Baduk or Weiqi). Among other board games I'm a fan of: backgammon, shogi, Blokus.

And I regularly post articles and videos on my blog ([neofanaro.io](http://neofanaro.io)) and YouTube channels, on Go, software, psychology, and a variety of different topics.

## 11 Personal Information

|                 |                                 |
|-----------------|---------------------------------|
| <b>Location</b> | São Paulo, São Paulo, Brazil    |
| <b>Phone</b>    | +55 (11) 97028-6739             |
| <b>Email</b>    | philippefanaro@gmail.com        |
| <b>Blog</b>     | neofanaro.io                    |
| <b>LinkedIn</b> | linkedin.com/in/philippe-fanaro |
| <b>GitHub</b>   | github.com/psygo                |